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Abstract:

This report discusses the detailed design of a Facial Smart Attendance and Lecture Recording System for the Richards Bay campus. The system was selected in Assignment 1 as the recommended solution to address the problem of manual attendance taking in large lecture classes. The report covers the full system configuration and components in the three subsystems, which consist of the Student Attendance Subsystem, the Lecturer Trigger and Timer Subsystem, and the Lecture Recording and Upload Subsystem. The integration of the new system with the university's existing facial scanning infrastructure is discussed in detail, along with the major components selected for each subsystem and their technical specifications. The report also addresses the embodiment design of the system, the integration plan for all lecture venues on campus, the modifications required to existing infrastructure, and the safety considerations and standards compliance to the system. The proposed design aims to automate attendance recording and lecture videoing in a way that is affordable, reliable and provides better services to staff and students.

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Introduction:

Universities nowadays are facing increasing pressure to reduce costs, improve energy efficiency and provide better environments for students and staff. Richards Bay campus faces these same challenges.

In Assignment 1, we were tasked to develop a Smart Campus system for Richards Bay campus. We identified the problems that Richards Bay campus is facing which is that manually taking attendance for large classes such as the first-year engineers, becomes tedious, unreliable and wastes time which can be spent teaching. The second is that energy is being unnecessarily wasted in lecture venues where lights and air conditioning are left on in empty rooms or are not adjusted based on the time of day or weather conditions outside, which increases the operational costs of the campus.

After discussing alternative solutions of smart campus systems that can be integrated to solve these problems and weighting them according to the Multi-Criteria Decision Making method, we determined that we had chosen the Facial Recognition Smart Attendance and Lecture Recording System as the recommended solution.

This report will focus on the detailed system design and concepts that need to be considered when designing a project.

1.) System Configuration and Components:

1.1. System Overview:

The recommended solution which is the Facial Smart Attendance and Lecture Recording System consists of three integrated systems that operate together. The three subsystems are:

- Subsystem 1- Student Attendance Subsystem: When students arrive at the lecture venue, each one scans their face as they enter the lecture room which automatically records their attendance.
- Subsystem 2- Lecturer trigger and Timer subsystem: When the lecturer scans their face, a timer of 5 minutes begins which is to allow the lecturer to setup for the lecture they will be teaching.
- Subsystem 3- Lecture Recording and Upload Subsystem: A camera at the back of the classroom automatically begins recording once the timer of 5 minutes ends, and when the lecture has finished, the recording is automatically uploaded to Moodle for students to access at any time.

1.2. Overall System block diagram:

FACIAL SMART ATTENDANCE AND LECTURE RECORDING SYSTEM - OVERALL BLOCK DIAGRAM

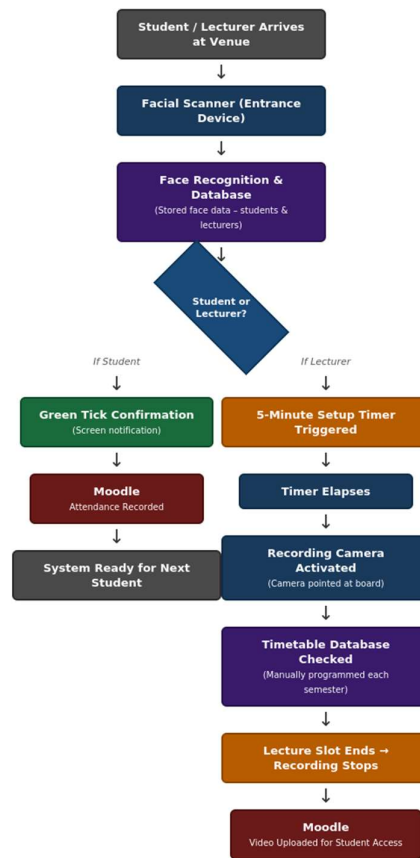


Figure 1: Block diagram of the whole system

1.3.) System Architecture and Integration with Existing Infrastructure:

The Facial Smart Attendance and Lecture Recording System will make use of the facial scanning devices that are already installed at the entrance of every lecture venue at the university. These devices already have built-in cameras and a database that stores the identity information of all students and lecturers, which means that the foundation of the system is already in place. The new system will build on this by adding:

- software integrated layer
- on-campus server
- PTZ camera
- wireless microphone to each venue.

When a person scans their face at the entrance, the existing facial scanner will verify their identity using OpenCV and FaceNet. This result is then sent to the on-campus server, which will determine whether the person is a student or a lecturer and respond accordingly. If it is a student, the server will communicate with Moodle to record their attendance for that lecture. If it is a lecturer, the

server will start a 5-minute timer to allow them to set up, then when timer has elapsed, the PTZ camera and wireless microphone will automatically begin recording the lecture. The server will also have a timetable programmed into it at the start of each semester, which it will use to determine when to stop the recording and upload it to Moodle. The only change needed to the existing infrastructure is adding the software integration layer to allow the facial scanner to communicate with the new local server. No physical changes to the existing hardware will be required.

1.4.) Major Components of each Subsystem:

Before selecting components, we must compare different types and see which fits the requirements best. The requirements are such that the components are:

- Affordable
- Reliable
- Easy to implement

1.4.1.) Component selection:

PTZ Camera:

Criteria:	HikvisionDS-2DE4425IWG-E	UNV 2MP LightHunter Mini PTZ	Axis P3245-V
Price:	R8500	R3200	R12000
Reliability:	High	High	High
Ease of implementation:	Moderate	Easy	Complex

The UNV 2MP LightHunter Mini PTZ was selected as it is the most affordable of the three options, is reliable and can run continuously, and is easy to implement as it connects directly to the existing LAN via a single PoE ethernet cable requiring no additional wiring.

Wireless Microphone:

Criteria:	Rode Wireless GO II	BOYA BY-WM4 Pro-K1	Sennheiser XSW-D
Price:	R6500	R1800	R9000
Reliability:	High	High	High
Ease of implementation:	Easy	Easy	Moderate

The BOYA BY-WM4 Pro-K1 was selected as it is significantly more affordable than the alternatives while still being reliable and providing sufficient range for any standard lecture venue. The receiver plugs directly into the server, making it very easy to implement.

Local Server:

Criteria:	Dell PowerEdge T40	HP ProLiant ML30 Gen10	Lenovo ThinkSystem ST50
Price:	R8500	R11000	R10200
Reliability:	High	High	High
Ease of implementation:	Easy	Moderate	Moderate

The Dell PowerEdge T40 was selected as it is the most affordable server of the three, is designed to run continuously without interruption making it highly reliable, and is straightforward to set up and configure compared to the alternatives.

Facial recognition Software:

Criteria:	OpenCV and FaceNet	Amazon Rekognition	Microsoft Azure Face API
Price:	Free	Pay per use	Pay per use
Reliability:	High	High	High
Ease of implementation:	Moderate	Easy	Easy

OpenCV and FaceNet were selected as they are completely free, very reliable, and run entirely on the local server meaning it doesn't depend on external services or internet connectivity. Although slightly more complex to set up than cloud-based alternatives, they are the only options that keep all biometric data on campus in line with POPIA requirements.

1.5) Technical Specifications:**1.5.1. Subsystem 1 – Student Attendance Subsystem:****University's Existing Facial Scanner:**

- Already placed at the entrance of every lecture venue with a built-in camera and database that contains all identity information of all registered students and lecturers to verify students and lecturer's identities.
- Has a built-in screen that displays confirmation messages to show successful verification, so students do not have to wait long before the next student is able to mark their attendance.

OpenCV and FaceNet:

- FaceNet converts a detected face into a unique numerical embedding that is compared against the stored identity information in the database to verify identity, with a

verification time of approximately 1–3 seconds per person depending on lighting and visibility [1][2].

1.5.2 Subsystem 2 – Lecturer Trigger and Timer Subsystem:

University's Existing Facial Scanner:

- When the scanner identifies the person scanning as a lecturer, it triggers the 5-minute setup timer instead of recording attendance to allow the lecturer to setup for the lecture.

Timer Software Module:

- Automatically triggered immediately when the lecturer's identity is verified and set to a time of 5 minutes, after which it sends a signal to the camera to begin recording without any manual input.

1.5.3 Subsystem 3 – Lecture Recording and Upload Subsystem:

UNV 2MP LightHunter Mini PTZ Camera:

- Records at Full HD 1080p with 4x optical zoom, allowing it to clearly capture the lecturer and any content written on the board
- Pan range of 350° and tilt range of 90°, meaning the camera can follow the lecturer as they move around the classroom

BOYA BY-WM4 Pro-K1 Wireless Microphone:

- Wireless transmission range of up to 60 metres with a stable connection, which is sufficient for any standard university lecture room no matter how far [4].
- Provides up to 6 hours of continuous battery life, enough battery to work consecutive lectures without needing to recharge [4].

Dell PowerEdge T40 Local Server:

- Powered by an Intel Xeon E-2224G processor with 8GB DDR4 RAM, providing enough processing power to simultaneously manage face recognition, the timer, recording triggers and Moodle uploads [3].
- Has 1TB of storage for timetable data, temporary videos and system logs, and designed to run continuously without interruption [3].

Moodle API (Upload):

- Automatically uploads the recording to the correct course page on Moodle using the timetable data to determine which course it belongs to.
- Students can then access the recording at any time through their Moodle account without any intervention from the lecturer.

1.6.) Embodiment Design:

1.6.1) Technical Design:

The physical placement of the new components is chosen after thorough consideration to ensure they work into the existing lecture room environment with minimal flaws. In our design we will place the PTZ camera at the back of the lecture room, we will mount it high up on the wall or ceiling which will give it a full view of the lecturer and the board at the front. Also this will see to it that the camera does not disrupt the student experience. We have a range of 2.5 to 3 meters for the mount height from the floor which we feel is just right to capture the full front of a standard university lecture room. Also we have chosen the BOYA BY-WM4 Pro-K1 which is a very small and light weight microphone which will clip on to the lecturer's clothing without adding any weight.

As for the receiver we will attach that to our local server which in turn takes care of the need for any visible cables in the class room. Also we will be using the Dell PowerEdge T40 which is a compact size server to be put in the university's present IT room on campus. This server will connect to all the lecture room cameras, present facial scanners and also Moodle through the university's present local area network (LAN) which means we do not have to add any major new cable infrastructure to the campus.

1.6.2.) Ergonomic Design:

The system has been designed to be simple and easy for both students and lecturers to use. Since the facial scanning devices are already installed at the entrance of every venue, students and lecturers won't need to learn how to use new hardware. The scanning process takes no intervention and takes around 3 seconds, which means that for a queue of students entering a lecture room can be done quickly without causing long delays at the entrance.

The existing facial scanner devices are already placed at a comfortable height at the venue entrances, which means they can be used by students of different heights. The confirmation screen on the scanner will display a clear green tick once a student's identity has been verified, giving immediate visual feedback so that the student knows they can continue into the room.

For the lecturer, the BOYA BY-WM4 Pro-K1 wireless microphone will be worn comfortably since it's small, lightweight and won't restrict their movement.

The 5-minute timer is triggered when the lecturer scans their face, as it gives the lecturer enough time to set up and prepare before the recording begins.

1.6.3.) Aesthetic Design:

The overall aesthetic goal of the system is to combine existing university environment with no extreme changes.

The existing facial scanner devices are already part of the university's infrastructure, so students and staff are already used to their appearance, therefore no changes will be made to these devices. The UNV 2MP LightHunter Mini PTZ camera has a compact and discreet design. When placed high on the back wall of the lecture room combined with its small shape also means it does not distract

students from focusing on the lecture. The camera's neutral colour also allows it to blend into its surroundings without standing out.

The BOYA BY-WM4 Pro-K1 wireless microphone transmitter is small enough that it is barely noticeable when clipped onto the lecturer's clothing. This is important because a visually bright microphone could make the lecture feel overly formal and recorded, which could negatively affect the natural flow of teaching.

1.6.4. Security and Data Privacy

Since the system collects and processes biometric data, data security is a very important consideration in the design. All biometric and attendance data is stored on the local on-campus server rather than a cloud-based server, so that the data never leaves the university's physical infrastructure and is not exposed to external threats.

Access to the server and its data will only be accessed by the authorised university IT staff only, so that student and lecturer data cannot be accessed without authorisation.

The system will also need to comply with South Africa's Protection of Personal Information Act (POPIA), which requires that personal data, including biometric data, is collected only for a specific purpose, stored securely and not shared without consent [5].

2. Integration with Existing Infrastructure

2.1 Integration Plan:

The Facial Smart Attendance and Lecture Recording System will be integrated into all lecture venues on the Richards Bay campus. Since the scanning devices are already installed at the entrance of every venue, the integration process will be focused on combining the scanners with new hardware's and software's.

Steps taken to integrate the new system:

Step 1: Server & Facial Scanner Setup:

- Install the local on-campus server in the university's existing IT room to act as the central hub of the system.
- Install the software integration layer into each of the facial scanners to allow communication with the server through the university's LAN.
- Program the timetable for all lectures in all the venues into the server at the start of each semester.

Step 2: Recording Equipment Installation:

- Install a PTZ camera and wireless microphone receiver in each of the lecture venues.
- Mount each PTZ camera at the back of the venue at a height around 2.5m and 3m.
- Connect each PTZ camera to the server using the existing LAN using a single ethernet cable.
- Connect each wireless microphone receiver to the server, which will be able to receive sound from the lecturer's transmitter.

Step 3: Configuration & Testing:

- Configure the server to communicate with Moodle, allowing automatic and immediate attendance to be recorded and video uploads for all of the venues.

- Test the full system in all the venues to ensure everything is functioning correctly.

2.2 Modifications and Upgrades to Existing Infrastructure:

We present a system that we designed to require minimal change to what is already in place at the universities which therefore keeps implementation costs low. We do have to do a few upgrades to make sure the system works well in all lecture venues. For the facial recognition we just need to install the software which in turn will get each scanner to talk to the campus server. As this is a software only change, we are still keeping the modification cost low. Also, we are looking at upgrading the network infrastructure, the university's Local Area Network, which we will need to put out more capacity to handle the large amount of video data which the PTZ cameras will be sending and uploading at the same time. The PTZ cameras use PoE which means they will get their power via the ethernet cable which they are plugged into therefore no separate power source is required. Therefore, we do not have to do any extra electrical work in the lecture rooms. The only power requirement is a small, dedicated power point for the Dell PowerEdge T40 server in the present IT or server room which is a very minor electrical add on that the university's facilities team can do at low cost.

3. Safety Considerations:

3.1 Safety Design Philosophies

In our design of the Facial Smart Attendance and Lecture Recording System, it is important to take note of possible failure modes and their consequences to understand what needs to be done to increase safety.

Some safety concerns include:

- Hardware failure: Hardware failures can happen if the PTZ camera, facial scanner, microphone, or server have faults. If the camera or microphone fails mid-lecture, the recording will be incomplete or become corrupt, meaning students lose access to that lecture on Moodle as well as possibly injure a student. If a facial scanner fails, students and lecturers will not be able to scan in, which means attendance cannot be recorded and the lecturer timer will not trigger. To reduce this risk, the IT department should perform regular hardware checks at the start of each semester, and a manual override option should be available so the lecturer can still start the recording if the scanner fails.
- Software failure: Software failures can take place in the facial recognition software or the timer module. If the recognition software crashes or has errors, students may be incorrectly identified or not identified at all, leading to false attendance. If the Moodle API fails, completed recordings or attendance data may not upload correctly, resulting in lost data. To reduce this risk, the server should be configured to automatically restart failed software services, and the IT department should perform regular software updates and testing.
- Data failure: Data failure could occur if attendance records or lecture recordings are lost due to server storage issues or corrupted files. Since all data is stored on the local on-campus server, a physical failure could result in permanent loss of biometric, attendance,

and recording data. To reduce this risk, the server should be configured to perform automatic daily backups to a secondary storage location on campus, ensuring data can be recovered if the primary storage fails.

3.2 Standards Compliance

In terms of design our system agrees with what the following key standards and regulations require:

POPIA (Protection of Personal Information Act, 2013):

- POPIA puts forward that biometric data including face scans is a special type of personal information which is to be processed under very strict legal terms.
- We have designed our system to comply with this by using facial data only for the attendance purposes. Also, we store all biometric data on the campus university server and do not put it out to external or cloud-based servers which is in agreement with POPIA's data minimization and purpose limitation rules.
- We only allow access to that stored biometric data to authorized IT personnel and we inform the students how we collect and use their data as required by POPIA's transparency rules.

IEC 62443:

- This is a set of standards which puts out requirements and processes for the implementation and maintenance of secure industrial automation and control systems and sets best practices for security which also has a system in place to assess the security performance.
- Our system does that through the use of HTTPS for all data transfer between the server and Moodle. Firewall protection is also implemented on the local server.
- Regular software updates which the IT department will handle will be done regularly to maintain compliance and protect against new vulnerabilities which goes in alignment with IEC 62443's continuous security maintenance requirement.

References:

[1] OpenCV, "Open Source Computer Vision Library," *opencv.org*. [Online]. Available: <https://opencv.org>. [Accessed: 2026].

[2] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Boston, MA, USA, Jun. 2015, pp. 815–823.

[3] Dell Technologies, "Dell EMC PowerEdge T40 Technical Specifications," *dell.com*. [Online]. Available: <https://www.dell.com/support/manuals/en-us/poweredge-t40>. [Accessed: 2026].

[4] BOYA, "BY-WM4 Pro Wireless Microphone System," *boyamic.com*. [Online]. Available: <https://store.boyamic.com/products/by-wm4-pro>. [Accessed: 2026].

[5] Republic of South Africa, *Protection of Personal Information Act (POPIA)*, Act No. 4 of 2013. Pretoria: Government Printer, 2013.

[6] South African Bureau of Standards, *SANS 10142-1: The Wiring of Premises — Part 1: Low-voltage Installations*, Ed. 1.8. Pretoria: SABS, 2012.